# Take-Home Assignment

# Senior Automation Infrastructure Engineer - Performance Investigation

The goal of this assignment is to evaluate how you approach investigating system performance as an Automation Infrastructure Engineer.

You will access our system **as a viewer** and analyze our product performance from a UI user point of view.

Your task is to develop a testing infrastructure and a test suite for analyzing UI performance and produce a performance analysis report based on the results.

We are interested in **your design solution, solution execution, framework flexibility and analytical insights.**

The assignment is designed to take around **4-5 hours** to complete.

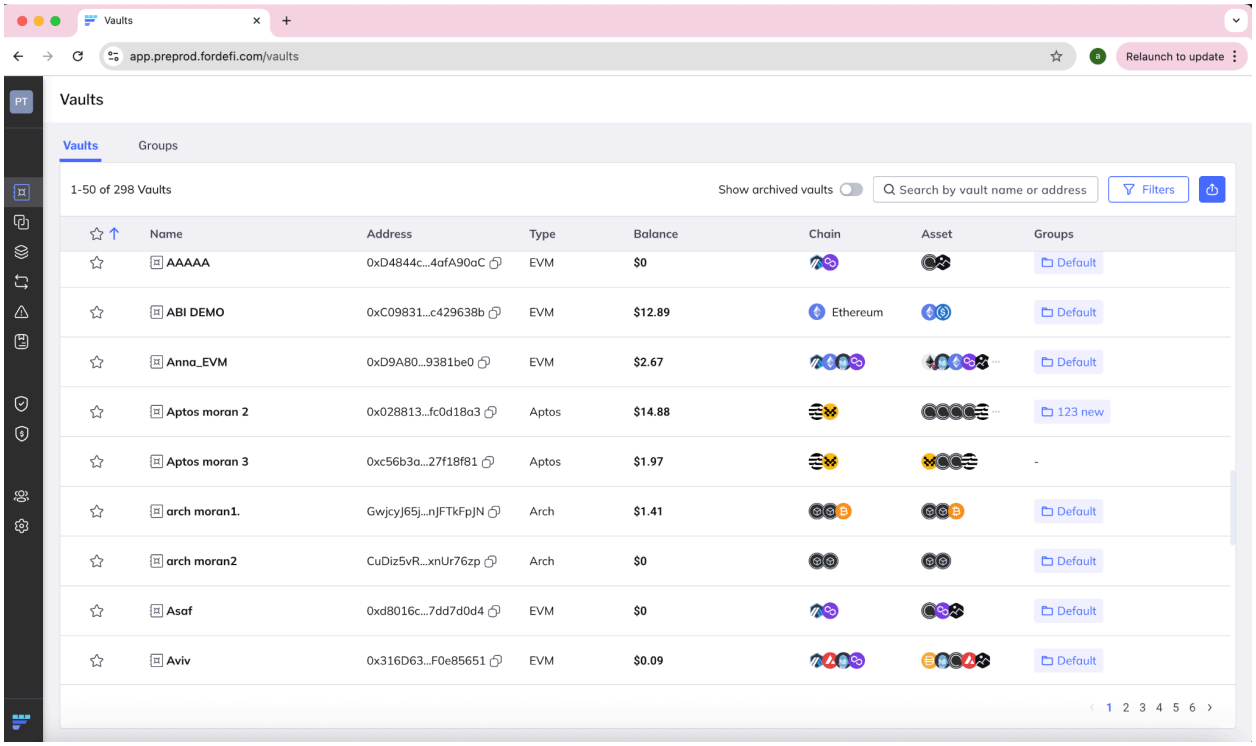
**Setup**

1. You will receive an **invitation email** to join the Fordefi system (If the email appears in your Spam folder, please access it from there)

2. Accept the invitation and follow the onboarding instructions.

3. Log in using: Google authentication, or username and password, and any additional authentication steps if required.

4. Onboarding is finished once you have access to the Fordefi website (URL https://app.preprod.fordefi.com/):




## Assignment

Once logged in, you will see multiple tabs and areas of the system.

Your goal is to **evaluate** the performance of the system from the perspective of a real viewer user, **investigate** the performance behavior of the system, and produce a performance investigation report.

You should:

1. Explore the system. Navigate through different parts of the product and identify areas that may have performance issues. Examples may include: Page load times, UI responsiveness, Network/API bottlenecks
2. Prioritize the findings. Identify the most critical performance problems, evaluate user impact and decide what should be investigated first.
3. Investigate the root causes. Choose a few poor-performing flows and try to pinpoint the bottlenecks
4. Ensure the testing framework is flexible enough to support new test cases easily
5. Analyze the results and generate a **Performance Investigation Report.**

**Submission**:

- Code
    - The test framework or scripts used to investigate the system
    - Any automation or performance tests you created
    - We are interested in seeing: How you structure the code, flexibility and maintainability of your solution, and how easily the solution could be extended.
- Report
    - Investigation methodology: how you approached the analysis and which tools/ techniques were used
    - Test scope / coverage: what areas, flows, or actions in the system were analyzed
    - Description of the issues
    - Prioritization of issues
    - Evidence and investigation details (screenshots, numeric metrics if available)
    - Possible root cause
    - Suggested improvement or solution
    - Include a brief explanation of how to add new test cases to the test suite
- *Note*: The report can be submitted as a document, presentation or any structured format you prefer.

If you have any questions during the assignment, feel free to ask.

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Good Luck! 🙂